Aditya Anil Tiwari

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WORK EXPERIENCE

Duralux Shades Inc — **Brooklyn, NY, USA**

Lead Mechanical Design Engineer — *Mar 2025 - Present*

- Led system-level mechanical architecture and performance trade studies for modular consumer-facing products, defining key design tradeoffs in cost, durability, manufacturability, and user experience while enabling scalable production of 5+ product variants.
- Reduced supplier-related rework by 15% by defining technical requirements, generating GD&T-controlled manufacturing drawings for sheet metal and plastic injection molding components, and approving pilot builds before full-scale production ramp-up.
- Executed complex 1D and 2D tolerance stack-up analyses (RSS and Monte Carlo simulations) on high-precision assemblies to ensure 99.7% (3-sigma) yield during high-volume production.

Mechanical Design Engineer — *Mar 2024 - Mar 2025*

- Delivered and launched 10+ production-ready mechanical products by owning concept development, SolidWorks CAD, rapid prototyping, validation, and production release, including 2 patent-pending mechanisms developed through DFMA and iterative testing.
- Reduced transport-related component failures by 20% by performing root-cause analysis on spring disengagement issues and redesigning mechanisms using robust snap-fit and gear-driven solutions, validated through functional and cycle testing.
- Managed technical communications and DFM reviews with international vendors, overseeing tooling validation and pilot builds to ensure design intent during high-volume production ramps while supporting ECOs and BOM revisions.

Worley India Pvt. Ltd. — **Mumbai, India**

Junior Project Manager — *Feb 2022 - Jun 2022*

- Improved project delivery timelines by 10% by coordinating engineering, procurement, and site teams through standardized documentation, technical reviews, and resolution of 25+ design-to-site discrepancies.
- Supported $1M+ piping and equipment procurement by reviewing material selection data, line designation tables, and hydrotest requirements, ensuring code-compliant, cost-effective technical sourcing decisions.

Junior Engineer, Stress and Materials Department — *Feb 2021 - Jan 2022*

- Ensured ASME B31.3 code compliance for 150+ piping systems by performing static and thermal stress analysis using CAESAR II, supporting long-term reliability across pressure vessels and rotating equipment.
- Increased stress analysis efficiency by 25% by standardizing calculation workflows, documentation, and critical line classification criteria, accelerating engineering decisions and project turnaround time.

DJS Racing – FSAE Team of DJ Sanghvi College of Engineering — **Mumbai, India**

Chassis Technical Lead | Technical Inspection Lead | Quiz Lead — *Mar 2017 - Jun 2020*

- Minimized risks and downtime by 15% while optimizing spare inventory and improving assembly efficiency by leading chassis system development from concept to manufacturing, implementing FMEA processes, and overseeing technical inspections.
- Achieved 30% weight reduction and 80% cost savings on bodywork and impact attenuators by refining composite designs and custom carbon fiber solutions validated via FEA - Abaqus simulations and physical crash tests.

SKILLS

- **Mechanical Design & CAD:** SolidWorks, Creo Parametric, AutoCAD, CATIA, Siemens NX, Autodesk Inventor, Fusion 360, MATLAB
- **Analysis & Simulation:** Ansys (Workbench, Mechanical, Composite), Abaqus, Inspire, HyperMesh, CAESAR II, NavisWorks, Python
- **Manufacturing & Prototyping:** Plastic Injection Molding, Extrusion, Sheet Metal, GD&T, CNC Machining - Router, Mill, Arc/TIG Welding, Fabric Welding, Composites - Layup & Vacuum Bagging, 3D Printing
- **Methods & Management:** DFMEA, DFM/DFA, Root Cause Analysis, ECOs, ERP/PLM, BOM, ISO 9001, Lean Manufacturing

PROJECTS

Sustain AI - Iterative AI for Sustainable Design Optimization, Columbia University — *Sep 2023 - Dec 2023*

- Developed an AI-assisted mechanical design optimization framework by integrating iterative feedback into existing CAD/CAE workflows, reducing design iterations and material rework by 40% during early-stage development.
- Enabled data-driven sustainability trade studies by evaluating eco-friendly material substitutions against mechanical performance constraints, improving early design decision quality and scalability across multiple product concepts.

Regis - Design and Fabrication of a Quadruped Robot, Columbia University — *Jan 2023 - May 2023*

- Designed and fabricated a 4-legged walking robot with Fusion360, leveraging FDM and SLS 3D printing techniques; employed Altair Inspire for topological optimization, attaining 12% weight savings in part design.
- Conducted URDF and GAIT analysis, resulting in a 15% increase in energy efficiency and 12% improvement in movement; ANSYS thermal to ensure heat performance and reliability.

EDUCATION

Columbia University | MS in Mechanical Engineering – GPA: 3.7/4 — **New York, NY, USA**

Robotics Club | Makerspace Member | Concentration – Robotics and Controls — *Aug 2022 - Dec 2023*

University of Mumbai | BE in Mechanical Engineering – GPA: 8.0/10 — **Mumbai, India**

SAE – Events Planning | Concentration – Automotive Design, Manufacturing, Project Management — *Jul 2016 - Oct 2020*